

Rongxuan Mu 穆荣轩👋

XR+AI HCI Designer / Creative Technologist / Indie Game Developer
Senior HCI Design Specialist @ PICO XR, Bytedance | +86 17612280645
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Professional Summary

XR+AI HCI Designer and Prototyper with proven experience in XR OS and games, grounded in robust HCI methodologies that span both hardware and software interaction frameworks. Leverages an architecture background to bring a refined spatial design intuition to immersive solutions. Passionate about pushing the boundaries of user-centric experiences and interactive environments.

Work Experiences

Jan 2024 - Now | Senior HCI Design Specialist @PICO XR, Bytedance

- Definition and prototyping of [PICO OS 6 & Project Swan](#) interaction framework, composing and coordinating variable interaction modalities, including eye gaze, hand tracking, speech, XR controller, ring, watch, and traditional pointer mouse with keyboard.
- Support specific HCI solutions for OS built-in Apps and the [Developer Platform](#).
- Lead and manage the prototyper team for quickly responsive soft/hardware prototypes and iterations.
- Host a series of internal sharing sessions, engaging 30+ industry pioneers.

June 2023 - Jan 2024 | HCI Designer and Prototyper @PICO XR, Bytedance

- Definition of multiple interaction techniques, delivering full-stack solutions from design details to prototypes and evaluations.

March 2022 - June 2023 | Technical Designer @PICO XR, Bytedance

- Designed and developed gameplay in some [XR game demos](#).

Aug 2021 - Feb 2022 | Real-estate Digital Product Manager Intern @DT-AIOT, Longfor Ltd

- Product management and UX design of the internal cross-platform [BIM apps](#), Interior Product Selection.

Skills

- Language: English (Proficient), Mandarin (Native)
- XR Prototyping: Prototype experiences on Quest, PICO, Apple Vision Pro, Magic Leap, Xreal, Rokid
- Programming: Unity C#, Python, Swift, Arduino
- Design Tools: Figma, Blender, Rhino+Grasshopper

Education

- Tianjin University - Bachelor's degree, Architecture
- University of California, LA - TJU+UCLA Joint Studio

Awards

- 2025 Spatial Joy - Silver Prize - [Geometria](#)
- 2024 Rokid Spatial Joy XR Competition - Bronze Prize, Best Practice of RTE - [Structuralism](#)
- 2024 H1 PICO Star Team - Star Team Contributor - PICO OS PanoScreen Team
- 2023 Rokid Developer Competition - Best Practice of Unity - [Muzuo XR](#)
- 2023 PICO Dev Jam - Silver Medalist - [Muzuo XR](#)
- 2023 Bytedance Staff Tech Jam - Silver Medalist, Best Business Potential - MR Interior Designer

Publications and Patents

- **Mu, R.**, Nie, Y., Cao, K., You, R., Wei, Y., & Tong, X. (2023). *Pilgrimage to Pureland: Art, Perception and the Wutai Mural VR Reconstruction*. International Journal of Human – Computer Interaction, 40(8), 2002 – 2018.
<https://doi.org/10.1080/10447318.2023.2271665>
- Han Rui & **Mu Rongxuan**. (2023). *From Architecture to Painting and Digital Media Art*. Time+Architecture (05), 36-41.
<https://doi.org/10.13717/j.cnki.ta.2023.05.03>
- *Memory Reconstruction: Design of a Rural Visitor Center*. (2023). China: Tianjin University Press. ISBN: 978-7-5618-7229-1
- 20+ Patents are not listed here according to NDA restrictions.

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个人概况

XR+AI 人机交互设计师和原型师，在XR操作系统和游戏领域拥有丰富的经验。具备扎实的人机交互方法论，涵盖硬件和软件交互框架。建筑学背景，为沉浸式解决方案带来精致的空间设计直觉。热衷于探索AI集成，以拓宽以用户为中心的体验和互动环境的边界。

工作经历

2024年1月 - 至今 | 资深人机交互设计专家 @PICO XR, 字节跳动

- 定义和原型化XR交互框架 ([Project Swan & PICO OS 6](#))，组合和协调多种交互模式，包括眼动追踪、手部追踪、语音、XR手柄、戒指、手表和传统的键鼠指针。
- 为操作系统内置应用和[PICO 开发者平台](#)提供特定的人机交互解决方案，例如拖放、旋转和缩放等基础交互事件细节。
- 领导和管理原型团队，支持内部软硬件原型快速迭代。
- 组织内部分享活动，涵盖30余位行业前沿嘉宾。

2023年6月 - 2024年1月 | 人机交互设计师和原型师 @PICO XR, 字节跳动

- 定义多项 XR 通用交互方式，交付从设计方案到原型工程的全链路解决方案，并负责端到端的长期 XR 交互性能评估测试。

2022年3月 - 2023年6月 | 技术策划 @PICO XR, 字节跳动

- 参与[多项 XR 游戏](#)原型策划和开发。

2021年8月 - 2022年2月 | 房地产数字化产品经理实习生 @DT-AIOT, 龙湖集团

- 负责内部跨平台BIM应用“室内产品选型”的产品管理和UX定义，参与敏捷开发管理流程。

通用技能

- 语言: 英语 (可作为工作语言), 普通话 (母语)
- XR原型开发: Quest, PICO, Apple Vision Pro, Magic Leap, Xreal, Rokid
- 编程: Unity C#, Python, Swift, Arduino
- 设计工具: Figma, Blender, Rhino+Grasshopper

教育背景

- 天津大学 - 建筑学学士学位
- 加州大学洛杉矶分校 (UCLA) - 天津大学+UCLA 联合工作室

奖项荣誉

- 2025 Spatial Joy XR开发大赛 - 银奖 - [Geometria](#)
- 2024 Rokid Spatial Joy XR大赛 - 铜奖, 最佳RTE实践奖 - [Structuralism](#)
- 2024 上半年 PICO 明星团队 - 明星团队贡献者 - PICO OS PanoScreen Team
- 2023 Rokid开发者大赛 - Unity最佳实践奖 - [Muzuo XR](#)
- 2023 PICO Dev Jam - 银奖 - [Muzuo XR](#)
- 2023 字节跳动 Staff Tech Jam - 银奖, 最佳商业潜力奖 - MR室内设计师

论文专利

- Mu, R., Nie, Y., Cao, K., You, R., Wei, Y., & Tong, X. (2023). *Pilgrimage to Pureland: Art, Perception and the Wutai Mural VR Reconstruction*. International Journal of Human – Computer Interaction, 40(8), 2002 – 2018.
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<https://doi.org/10.13717/j.cnki.ta.2023.05.03>
- Memory Reconstruction: Design of a Rural Visitor Center*. (2023). China: Tianjin University Press. ISBN: 978-7-5618-7229-1
- 根据保密协议限制，20余项专利暂未列出